

PARSA AHMADI

✉ p3ahmadi@uwaterloo.com  linkedin.com/in/parsa-ahmadi2006  github.com/ParsaA2006  Personal Website

Education

University of Waterloo

Expected Graduation: Apr 2029

Bachelor of Applied Science in Mechatronics Engineering | Minor Artificial Intelligence

Waterloo, ON

- Coursework: Data Structures and Algorithms, Digital Logic | **GPA: 4.0/4.0**

Technical Skills

Languages: Python, TypeScript, JavaScript, C, C++, C#, Java, SQL, HTML/CSS

Developer Tools: Azure, Git, AWS, Docker, Postman, PostgreSQL, SQL Server, MySQL, Redis, Jupyter Notebook, Jira

Frameworks/Libraries: React, Next.js, Angular, Blazor, ASP.NET, .NET, Entity Framework, Node.js, Express, PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Pandas, OpenCV, ROS2, PyQt

Experience

BTNX

Sep 2025 – Dec 2025

Software Engineering Intern

Toronto, ON

- Developed a **full-stack ERP SaaS platform** using **React**, **Typescript**, and **.NET**, building **30+** end-to-end pages and designing the normalized SQL Server schema, serving **1,000+** active users
- Reduced load times by **30%** across key modules through optimizing **LINQ** queries in **C#**, implementing **async/await** validation, and improving **SQL** performance by introducing table indexes, stored procedures, and a Redis lookup cache
- Engineered scalable **RESTful APIs** with **ASP.NET** and **Entity Framework** to handle **5k+** **daily requests**, reducing downtime by **20%** and implementing integrations with Microsoft Graph and Amazon SP API
- Built an AI-powered conference tracking module with **Blazor** and **Python**, training OCR and NLP models using **PyTorch** to extract business card data and auto-populate attendee profiles with **96%** accuracy
- Developed **Python scripts** to train and deploy an **XGBoost** classification model for diagnostic strips with ONNX Runtime inference integrated into a **Blazor** interface, achieving **99%** strip reader accuracy

Neurosnap

May 2025 – Aug 2025

Software Engineering Intern

Toronto, ON

- Built responsive research dashboards for AI-driven protein and enzyme analysis using **TypeScript** and **React**, reducing dashboard render time by **35%** for large bioinformatics datasets
- Built backend services in **Python** using **Flask**, integrating **ML pipelines** for molecular docking and sequence analysis, and reducing processing errors by more than **25%**
- Designed **PostgreSQL** schemas and indexed queries for storing protein structures, variant predictions, and experiment results, improving query performance by nearly **50%**
- Integrated **RESTful APIs** with **Python** and **Flask** for NeuroFold model inference and containerized batch prediction pipelines with **Docker**, automating research and saving laboratories **\$20000+** annually

Waterloo Aerial Robotics Group (WARG)

May 2025 – Dec 2025

Autonomy Team Member

Waterloo, ON

- Developed computer vision modules for landing pad detection and obstacle avoidance, increasing detection accuracy to **90%** by training ML models with **OpenCV**, **TensorFlow**, and deploying inference pipelines in **C++** within **ROS2**
- Implemented real-time telemetry features in the IMACS 2.0 ground-station, reducing communication latency from **150 ms** to **90 ms** by optimizing asynchronous data handling using **Python**, **PyQt**, and **ROS2**

Projects

Premier League Predictor | Python, FastAPI, React, TypeScript, Docker

Nov 2025

- Created a model to predict match outcomes, integrating **data scraping** with feature engineering and machine learning using **XGBoost** and **scikit-learn**
- Containerized the full-stack application using **Docker** with multi-stage builds and integrated **CI/CD** pipelines, enabling automated testing, consistent deployments across environments, and reducing setup time by over **80%**

Tic-Tac-Toe Solver Robot | C++, RobotC

Dec 2024

- Programmed an autonomous LEGO EV3 Tic-Tac-Toe robot in **C++**, implementing color-sensor board detection and real-time game-state processing to interpret human moves
- Integrated and calibrated the EV3's **multi-axis motors** by mapping grid coordinates to calibrated motor rotation angles, ensuring accurate and consistent move placement across the 3×3 grid

UNIVERSITY OF WATERLOO

UNOFFICIAL GRADE REPORT

Parsa Ahmadi
21131398

2A Mechatronics Engineering, Honours, Co-operative Program

Winter 2026

MSE	442	Impact Info Sys on Orgs & Soc
MTE	182	Physics 2: Dynamics
MTE	200A	Seminar
MTE	201	Exp Measure & Stat Analysis
MTE	202	Ordinary Differential Equation
MTE	219	Mechanics of Deformable Solids
MTE	262	Intro to Digital Logic

Term Average: N/A

Decision:

Fall 2025

COOP	2	Co-operative Work Term	CR
PD	20	Strategies for Career Success	CR

Term Average: N/A

Decision:

Spring 2025

MATH	118	Calculus 2 (Eng)	89
MTE	100B	Seminar	
GENE	120	First-Year Eng Seminar	
MTE	111	Material Struct & Props	92
MTE	140	Algorithms & Data Structures	93
MTE	120	Circuits	88
MTE	119	Statics	94

Term Average: 90.91

Decision: Excellent Standing

Winter 2025

PD	19	Tactics for Workplace Success	CR
COOP	1	Co-operative Work Term	CR

Term Average: N/A

Decision:

Fall 2024

CHE	102	Chemistry for Engineers	90
GENE	119	First-Year Engineering Seminar	
MTE	121	Digital Computation	91
MTE	100	Mechatronics Engineering	95
MATH	116	Calculus 1 (Eng)	100
MATH	115	Linear Algebra (Eng)	97

Term Average: 94.64

Decision: Excellent Standing